

Split and Merge (SaM): Efficient Change-Point Detection with Additive Computational Cost in Low-Rank Matrices

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Abstract: High-dimensional matrix-valued data often exhibit latent low-rank structures that evolve over time and experience abrupt changes. Detecting multiple change points in such settings is statistically and computationally challenging, since conventional approaches repeatedly solve nuclear norm regularized estimation problems over dense candidate grids. In this paper, we study a trace regression model with time-varying low-rank coefficient matrices and propose a Split-and-Merge (SaM) framework that separates coefficient estimation from change-point localization. SaM first estimates low-rank parameters on a coarse segmentation, and then identifies change points via an ℓ_0 -regularized optimization without further parameter re-estimation. This strategy reduces the computational complexity from multiplicative to additive in the grid size. We establish sharp localization guarantees under a flexible near low-rank regime. Extensive simulation studies demonstrate that SaM is highly competitive in terms of both statistical accuracy and computational efficiency. Across real-data analyses from environmental, financial, and transportation network domains, the proposed

method identifies interpretable change points associated with structural shifts in underlying low-rank matrices.

Key words and phrases: Trace regression, near low-rank, change-point, dynamic programming, high-dimensional data

1. Introduction

Many real-world high-dimensional datasets exhibit intrinsic low-rank structures that capture their dominant patterns and dependencies. In practice, observations are often incomplete, noisy, or corrupted, which degrades the performance of classical estimation methods. Low-rank matrix recovery offers a principled approach for reconstructing such latent structures from partial or contaminated measurements and has become a central topic in modern high-dimensional statistics and machine learning.

Despite substantial progress in static settings, dynamic low-rank structures in which the underlying matrix evolves over time and may undergo abrupt structural changes remain relatively underexplored. Existing work has mainly focused on vector autoregressive (VAR) models. For instance, Bai et al. (2020) studied a VAR model with a regression matrix decomposed into a stable low-rank component and a time-varying sparse component, while Bai et al. (2023) extended this framework to allow both dynamic low-rank and sparse structures using rolling-window strategies for

multiple change-point detection. Related information-criterion-based approaches were considered in Enikeeva et al. (2025). More recently, Shi et al. (2024) moved beyond VAR models and investigated a general trace regression framework for dynamic low-rank matrix recovery.

To describe the data with latent dynamic low-rank structure, we study the trace regression model in this paper. Our setting is not restricted to a specific VAR problem, but is formulated under a general trace regression framework, which unifies a broad class of models including linear regression, multivariate regression, compressed sensing, and matrix completion. Trace regression is a fundamental model that continues to attract significant attention in modern machine learning (Jang et al., 2024). Consider a sequence of observations $\mathcal{D} = \{(y_i, \mathbf{X}_i)\}_{i=1}^n$, where $y_i \in \mathbb{R}$ denotes the response variable and $\mathbf{X}_i \in \mathbb{R}^{d_1 \times d_2}$ is the associated matrix-valued covariate. Let $[x] := \{1, 2, \dots, x\}$ for any integer $x > 0$. The data is assumed to be generated from the following piecewise trace regression model:

$$y_i = \langle \mathbf{X}_i, \boldsymbol{\theta}_j^* \rangle + \varepsilon_i, \quad \text{for } \tau_{j-1}^* < i \leq \tau_j^*, \quad (1.1)$$

where $i \in [n]$ indexes the time of the observed data-point, $\varepsilon_i \in \mathbb{R}$ denotes a random noise term, and $j \in [\kappa^* + 1]$ indexes the segment $(\tau_{j-1}^*, \tau_j^*]$ separated by the change-points $\mathcal{T}_{\kappa^*}^* := \{\tau_j^*\}_{j=1}^{\kappa^*}$, with the convention $\tau_0^* = 0$ and $\tau_{\kappa^*+1}^* = n$. The matrix-valued parameter $\boldsymbol{\theta}_j^* \in \mathbb{R}^{d_1 \times d_2}$ is piecewise constant

across segments, with $\boldsymbol{\theta}_j^* \neq \boldsymbol{\theta}_{j+1}^*$ for all $j \in [\kappa^*]$. Define the local spacing parameter and local jump size parameter as

$$\delta_j = \tau_j^* - \tau_{j-1}^*, \text{ for } j \in [\kappa^* + 1]; \quad \Xi_j = \|\boldsymbol{\theta}_{j+1}^* - \boldsymbol{\theta}_j^*\|_F, \text{ for } j \in [\kappa^*]. \quad (1.2)$$

Furthermore, we denote $\delta_{\min} = \min_{j \in [\kappa^* + 1]} \delta_j$ and $\Xi_{\min} = \min_{j \in [\kappa^* + 1]} \Xi_j$. In this framework, both the number of change-points κ^* and the dimensions d_1, d_2 are allowed to grow with the sample size n .

In the multi-change-point setting, most existing methods discretize time into candidate locations and repeatedly solve low-rank estimation problems on the induced segments (Negahban and Wainwright, 2011). In high-dimensional regimes with large (d_1, d_2) , however, even a *single* nuclear-norm-based low-rank estimation is computationally expensive, with complexity $\text{Nuclear}(d_1, d_2, n) = O(d_1 d_2 n + d_1 d_2 (d_1 \vee d_2))$ (Shi et al., 2024). Re-estimating parameters over many candidate segments therefore becomes prohibitive as both the sample size and matrix dimensions grow, while dense candidate grids introduce substantial redundancy through highly overlapping segments and nearly identical solutions. To reduce this burden, several recent works adopt non-full grid search strategies in change-point analysis Killick et al. (2012); Li et al. (2023); Kovács et al. (2024); Cho and Owens (2024). Such approaches, however, inherently trade statistical accuracy for computational efficiency: restricting the search to $\mathcal{B}_n$ candidates

yields at best an $O_p(n/\mathcal{B}_n)$ localization rate, whereas achieving high statistical precision requires enlarging $\mathcal{B}_n$, leading to a computational cost of $O(\mathcal{B}_n \cdot \text{Nuclear}(d_1, d_2, n))$ that is again infeasible in high dimensions.

To overcome these challenges, we propose a *Split-and-Merge* (SaM) framework that decouples low-rank coefficient estimation from change-point localization. By separating parameter learning and location search, the proposed method enables accurate detection of multiple change points with substantially reduced computational cost. In the *split* step, we start from a moderately large set of coarse, well-separated initial change-point candidates (hereafter referred to as *initializers*). These candidates induce a segmentation of the data, on which we estimate low-rank coefficient matrices only once. In the *merge* step, we keep these initial estimates fixed and select the final change points by solving a discrete optimization problem, without any further coefficient re-estimation.

Contributions. The main contributions of the proposed *Split-and-Merge* (SaM) framework are summarized as follows. We introduce a novel decoupling strategy that separates parameter learning from location search, transforming the computational cost of multiple change-point estimation from multiplicative to additive, and to the best of our knowledge, SaM achieves the fastest runtime under this high-dimensional low-rank setting

with sub-second performance in practice. Under a near-low-rank multiple change-point model, SaM attains near-optimal localization accuracy, and with only one additional global refinement step (Appendix ??), it further achieves an improved convergence rate. For practical implementation, we develop a new cross-validation procedure for tuning parameter selection, which is enabled by the extremely low per-run cost of SaM and allows flexible tuning without additional computational burden. We also propose an adaptive version of SaM for automatically constructing suitable initializers. Extensive simulation studies demonstrate that SaM substantially outperforms existing baselines in both statistical accuracy and computational efficiency, and three real-data applications further illustrate the practical usefulness of the proposed method, with additional simulations, algorithmic details, and theoretical proofs deferred to the Appendix.

Related Literature. Most existing methods for multiple change-point detection can be broadly divided into two paradigms: CUSUM-based procedures and loss-based optimization approaches. CUSUM-type statistics form the classical foundation of change-point analysis due to their strong theoretical guarantees and computational efficiency (Csörgö and Horváth, 1997). By combining CUSUM with recursive schemes such as binary segmentation (Scott and Knott, 1974) and its modern variants (Fryzlewicz,

2014; Baranowski et al., 2019; Kovács et al., 2023), these methods extend naturally to multiple change-point settings. However, in high-dimensional regimes, CUSUM-based approaches are largely model-specific and primarily focused on mean-shift problems, with only limited extensions to more complex models (e.g., Cho et al., 2025; Liu et al., 2020). Moreover, constructions based on high-dimensional estimators such as Lasso further increase computational cost (Wang et al., 2021; Liu et al., 2026), making it difficult to scale these methods to large dimensions. An alternative line of work formulates change-point detection as a loss-based optimization problem. Fused-type Lasso methods (Rinaldo, 2009) are computationally attractive but tend to overestimate the number of change points, requiring additional screening via information criteria (Bai et al., 2020; Safikhani and Shojaie, 2022; Bai et al., 2023). In contrast, ℓ_0 -penalized likelihood approaches (Killick et al., 2012; Leonardi and Bühlmann, 2016; Yang et al., 2025) achieve sharp localization guarantees and often attain optimal rates after refinement (Wang et al., 2023; Xu et al., 2024; Wang et al., 2025).

Notation. For any integer $x > 0$, let $[x] := \{1, \dots, x\}$ and $[x]^- := \{x, \dots, 1\}$. For asymptotics, we denote $x_n \lesssim y_n$ or $x_n = O(y_n)$ if $\forall n$, $x_n \leq c_1 y_n$ for some universal constant $c_1 > 0$. $a_n = o(b_n)$ means $a_n/b_n \rightarrow 0$ as $n \rightarrow \infty$, and $X_n = o_p(Y_n)$ if $X_n/Y_n \rightarrow 0$ in probability. We call a

positive sequence $\{a_n\}_{n \in \mathbb{Z}^+}$ a diverging sequence if $a_n \rightarrow \infty$ as $n \rightarrow \infty$. For a matrix $\mathbf{X} \in \mathbb{R}^{d_1 \times d_2}$, $\|\mathbf{X}\|_{op}$, $\|\mathbf{X}\|_F$, and $\|\mathbf{X}\|_*$ denote the operator, Frobenius, and nuclear norms, respectively. We define $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{A}^\top \mathbf{B})$ and $\text{vec}(\mathbf{X})$ as the vectorization of $\mathbf{X}$. Let $\rho_l(\mathbf{X})$ be the l -th largest singular value. For a differentiable function L , ∇L denotes its gradient, and we set $d = \min(d_1, d_2)$. For a finite set $\mathcal{I}$, $|\mathcal{I}|$ denotes its cardinality. We also let $\mathbf{I}_d$ denote the d -dimensional identity matrix. For any integer $\kappa \geq 1$, define the full parameter matrix $\boldsymbol{\theta}_{[\kappa+1]} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_{\kappa+1}) \in \mathbb{R}^{d_1 \times d_2(\kappa+1)}$ with each given coefficient matrix $\boldsymbol{\theta}_j \in \mathbb{R}^{d_1 \times d_2}$ for $j \in [\kappa+1]$. In particular, the true full parameter matrix is denoted by $\boldsymbol{\theta}_{[\kappa^*+1]}^* = (\boldsymbol{\theta}_1^*, \dots, \boldsymbol{\theta}_{\kappa^*+1}^*) \in \mathbb{R}^{d_1 \times d_2(\kappa^*+1)}$. Let $v \in [n]$ be the right boundary of a segment and $\kappa < v$ be the number of potential change-points. We denote the set of strictly ordered candidate locations as $\mathbb{T}_\kappa(v) := \{\{\tau_j\}_{j=1}^\kappa | 0 = \tau_0 < \tau_1 < \dots < \tau_{\kappa+1} = v\}$. Its relaxation, which permits overlapping locations, is defined as $\mathbb{U}_\kappa(v) := \{\{u_j\}_{j=1}^\kappa | 0 = u_0 \leq u_1 \leq \dots \leq u_{\kappa+1} = v\}$.

2. Methodology

This section introduces the SaM framework and evaluates its computational complexity. We begin by defining the coefficient estimators. Let the dataset $\mathcal{D}$ follow the distributional sequence in (1.1). For any segment

$(u, v]$ with $1 \leq u < v \leq n$, the cumulative loss for a common parameter $\boldsymbol{\theta} \in \mathbb{R}^{d_1 \times d_2}$ is defined as $\mathcal{L}(u, v; \boldsymbol{\theta}) = \sum_{i=u+1}^v (y_i - \langle \mathbf{X}_i, \boldsymbol{\theta} \rangle)^2$. The corresponding penalized estimator $\check{\boldsymbol{\theta}}_{(u,v]}^\lambda$ is obtained via:

$$\check{\boldsymbol{\theta}}_{(u,v]}^\lambda = \underset{\boldsymbol{\theta} \in \mathbb{R}^{d_1 \times d_2}}{\operatorname{argmin}} \left\{ \mathcal{L}(u, v; \boldsymbol{\theta}) + \lambda \sqrt{v-u} \|\boldsymbol{\theta}\|_* \right\}. \quad (2.3)$$

Here, the scaling term $\sqrt{v-u}$ adaptively normalizes the penalty for different interval sizes, while the nuclear norm encourages the low-rankness of the parameter matrix.

We next define the total loss. For any $\kappa \geq 0$ and integer $v \in [n]$, given $\mathcal{U} \in \mathbb{U}_\kappa(v)$ and $\boldsymbol{\theta}_{[\kappa+1]}$, the total loss over all segments is defined as

$$\text{TL}(\mathcal{U}, \boldsymbol{\theta}_{[\kappa+1]}) := \sum_{j=1}^{\kappa+1} \mathcal{L}(u_{j-1}, u_j; \boldsymbol{\theta}_j). \quad (2.4)$$

Analogously, for any locations $\mathcal{T} \in \mathbb{T}_\kappa(v)$, we denote by $\text{TL}(\mathcal{T}, \boldsymbol{\theta}_{[\kappa+1]})$ the total loss induced by the locations $\mathcal{T}$.

2.1 The Algorithm SaM

Overview of SaM. Our proposed framework is built upon the SaM algorithm, which consists of two steps: a *split step* and a *merge step*; see Algorithm 1. Figure 1 provides a concrete illustrative example.

In the *split step*, we start from a collection of initializers $\tilde{\mathcal{T}}_\kappa$ that satisfy relatively mild conditions—specifically, the number of initializers is suffi-

Algorithm 1 Split and Merge SaM($\mathcal{D}, \tilde{\kappa}, \lambda, \gamma$)

Input: Data $\mathcal{D}$, number $\tilde{\kappa}$, tuning parameters $\lambda, \gamma > 0$.

Output: The proxy change-point estimators $\hat{\mathcal{T}}_{\tilde{\kappa}}$

- 1: Set suitable initializers $\tilde{\mathcal{T}}_{\tilde{\kappa}} = \{\tilde{\tau}_j\}_{j=1}^{\tilde{\kappa}} \in \mathbb{T}_{\tilde{\kappa}}(n)$.
 - 2: (Split Step) Compute initial coefficients $\tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]} := (\check{\boldsymbol{\theta}}_{(0, \tilde{\tau}_1]}^{\lambda}, \dots, \check{\boldsymbol{\theta}}_{(\tilde{\tau}_{\tilde{\kappa}}, n]}^{\lambda})$ by solving (2.3) with $\tilde{\mathcal{T}}_{\tilde{\kappa}}$ and λ .
 - 3: (Merge Step) Compute proxy estimator $\hat{\mathcal{T}}_{\tilde{\kappa}}$ using MergeDP($\mathcal{D}, \tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}, \gamma$) in Algorithm 2.
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ciently large and their spacing is sufficiently wide (see Assumption 1). Given these initializers, we estimate $\tilde{\kappa} + 1$ coefficient matrices on the segments induced by the initializers by solving the nuclear-penalized objective (2.3). This yields the initial parameter estimates

$$\tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]} = (\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{\tilde{\kappa}+1}) := (\check{\boldsymbol{\theta}}_{(0, \tilde{\tau}_1]}^{\lambda}, \dots, \check{\boldsymbol{\theta}}_{(\tilde{\tau}_{\tilde{\kappa}}, n]}^{\lambda}).$$

This step only requires solving the nuclear-penalized problem $\tilde{\kappa} + 1$ times. Under the regime $\tilde{\kappa} \ll n \vee d$, the computational complexity of the split step is $O(\tilde{\kappa} \cdot \text{Nuclear}(d_1, d_2, n))$.

In the *merge step*, no additional coefficient estimation is required. Instead, we directly plug the initial estimates $\tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}$ into an ℓ_0 -regularized objective introduced in the next section and compute its optimal solution.

After a simple deduplication operation (2.8), this solution yields a proxy estimator of the change-point set, denoted by $\widehat{\mathcal{T}}_{\tilde{\kappa}}$. The resulting ℓ_0 -regularized problem can be efficiently solved via a dynamic programming strategy analogous to the classical segment neighborhood (SN) approach (Auger and Lawrence, 1989), which we refer to as *merged dynamic programming*.

Since the merge step avoids any further coefficient estimation and relies solely on MergeDP, its computational complexity is $O(n^2)$. Consequently, the overall computational complexity of the SaM algorithm is additive, i.e., $O(\tilde{\kappa} \cdot \text{Nuclear}(d_1, d_2, n) + n^2)$.

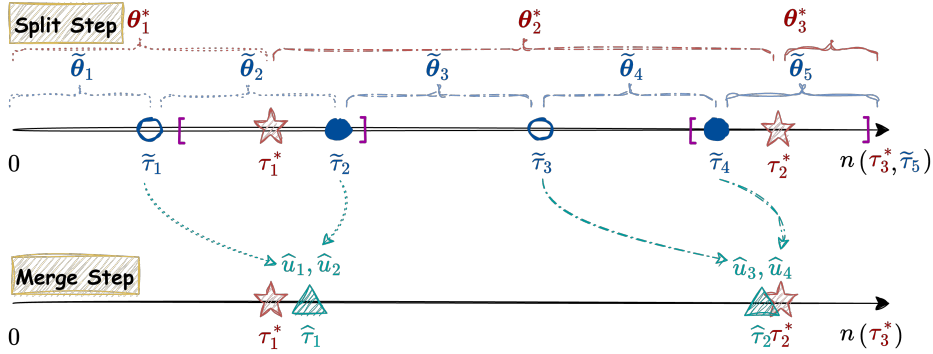


Figure 1: Illustration of Algorithm 1. Red stars, blue circles, and cyan triangles denote τ_2^* , $\tilde{\tau}_4$, and $\hat{\tau}_2$, respectively; blue and red braces indicate $\tilde{\theta}_{[5]}$ and $\theta_{[3]}^*$.

An ℓ_0 -regularized Objective. Given the $\tilde{\kappa} + 1$ initial coefficients $\tilde{\theta}_{[\tilde{\kappa}+1]}$, we globally reassign $\tilde{\kappa}$ candidate locations by solving the following ℓ_0 -

regularized optimization problem:

$$\widehat{\mathcal{U}}_{\tilde{\kappa}} = \operatorname{argmin}_{\mathcal{U} \in \mathbb{U}_{\tilde{\kappa}}(n)} \left\{ \text{TL}(\mathcal{U}, \tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}) + \gamma \sum_{\tilde{j}=1}^{\tilde{\kappa}+1} \mathbf{1}_{\{u_{\tilde{j}-1} < u_{\tilde{j}}\}} \right\}, \quad (2.5)$$

where $\gamma > 0$ controls the effective number of intervals. The solution $\widehat{\mathcal{U}}_{\tilde{\kappa}}$ is referred to as the *raw estimated locations*. Notably, the optimization problem (2.5) is solved over the relaxed space $\mathbb{U}_{\tilde{\kappa}}(n)$, which allows adjacent locations to *coincide*. Consequently, the raw estimated locations may contain duplicates. Removing such duplicates yields the proxy change-point estimator $\widehat{\mathcal{T}}_{\tilde{\kappa}}$. To formalize this operation, we define a deduplication operator for any $\mathcal{U} \in \mathbb{U}_{\kappa}(v)$ with $\kappa \in [\tilde{\kappa}]$ and $v \in [n]$ as

$$\text{Unique}(\mathcal{U}) = \{u_{\tilde{j}} \mid u_{\tilde{j}} \neq u_{\tilde{j}+1}, \tilde{j} \in [k]\}. \quad (2.6)$$

That is, consecutive duplicate locations are collapsed, and the endpoint v is excluded. The proxy change-point estimator is then defined as

$$\widehat{\mathcal{T}}_{\tilde{\kappa}} = \text{Unique}(\widehat{\mathcal{U}}_{\tilde{\kappa}}), \text{ and } \widehat{\kappa} = |\text{Unique}(\widehat{\mathcal{U}}_{\tilde{\kappa}})|.$$

Solving (2.5) is challenging due to its discrete, non-smooth, and non-convex nature. The solution space grows exponentially with $n^{\tilde{\kappa}}$ and may contain multiple global optima, rendering naive search strategies computationally infeasible. Moreover, applying stochastic optimization methods Kaul et al. (2019) to solve the ℓ_0 -regularized objective (2.5) is often problematic: their performance is highly sensitive to hyperparameter choices and

they typically incur substantial computational overhead in high-dimensional settings. These issues motivate the need for an efficient and globally optimal procedure tailored to the proposed ℓ_0 formulation.

Merged Dynamic Programming. To efficiently solve the ℓ_0 -regularized objective (2.5), we exploit its additive structure via dynamic programming. Specifically, for any $v \in [n]$ and $\kappa \in [\tilde{\kappa}]$, given the data $\mathcal{D}_0^v := \{(y_i, \mathbf{X}_i)\}_{i=1}^v$ and initial coefficients $\tilde{\boldsymbol{\theta}}_{[\kappa+1]}$, we define the Bellman function

$$\mathcal{F}(v; \tilde{\boldsymbol{\theta}}_{[\kappa+1]}) := \min_{u \in \mathbb{U}_\kappa(v)} \left\{ \sum_{\tilde{j}=1}^{\kappa+1} [\mathcal{L}(u_{\tilde{j}-1}, u_{\tilde{j}}; \tilde{\boldsymbol{\theta}}_{\tilde{j}}) + \gamma \mathbf{1}_{\{u_{\tilde{j}-1} < u_{\tilde{j}}\}}] \right\}. \quad (2.7)$$

By construction, $\mathcal{F}(n; \tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]})$ equals the minimum value of (2.5). Moreover, $\mathcal{F}$ admits the following recursion for $v \in [n]$:

$$\mathcal{F}(v; \tilde{\boldsymbol{\theta}}_{[\kappa+1]}) = \min_{u \leq v} \left\{ \mathcal{F}(u; \tilde{\boldsymbol{\theta}}_{[\kappa]}) + \mathcal{L}(u, v; \tilde{\boldsymbol{\theta}}_{\kappa+1}) + \gamma \mathbf{1}_{\{u < v\}} \right\}, \kappa \in [\tilde{\kappa}], \quad (2.8)$$

with first part $\mathcal{F}(v; \tilde{\boldsymbol{\theta}}_{[1]}) = \mathcal{L}(0, v; \tilde{\boldsymbol{\theta}}_1) + \gamma$. The raw estimated locations $\hat{\mathbf{u}}_{\tilde{\kappa}}$ are recovered by standard backtracking from (2.8), and subsequently deduplicated via the operator (2.6) to obtain the proxy change-point estimators $\hat{\mathcal{T}}_{\tilde{\kappa}}$. The complete procedure is summarized in Algorithm 2. For the nested loops in Lines 2–6 of Algorithm 2, we provide a more memory-efficient implementation in Appendix ?? (see Algorithm ??).

2.2 Practical Implementation and Data-Driven Method

Algorithm 2 Merged Dynamic Programming $\text{MergeDP}(\mathcal{D}, \tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}, \gamma)$

Input: Data $\mathcal{D}$, initial coefficients $\tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}$, tuning $\gamma > 0$.

Output: The proxy change-point estimators $\hat{\mathcal{T}}_{\tilde{\kappa}}$

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1: Set  $\mathcal{U} = \emptyset$ ,  $\mathbf{P} \leftarrow (0) \in \mathbb{R}^{\tilde{\kappa} \times n}$ .

2: for  $\kappa \in [\tilde{\kappa}]$  do

3:   for  $v \in [n]$  do

4:      $[\mathbf{P}]_{\kappa, v} \leftarrow \underset{u \leq v}{\operatorname{argmin}} \left\{ \mathcal{F}(u; \tilde{\boldsymbol{\theta}}_{[\kappa]}) + \mathcal{L}(u, v; \tilde{\boldsymbol{\theta}}_{\kappa+1}) + \gamma \mathbf{1}_{\{u < v\}} \right\}$ 

5:   end for

6: end for

7: Set  $(\kappa, v) \leftarrow (\tilde{\kappa}, n)$ 

8: while  $\kappa > 0$  do

9:    $\mathcal{U} \leftarrow \mathcal{U} \cup \{[\mathbf{P}]_{\kappa, v}\}; (\kappa, v) \leftarrow (\kappa - 1, [\mathbf{P}]_{\kappa, v})$ 

10: end while

11:  $\hat{\mathcal{T}}_{\tilde{\kappa}} \leftarrow \text{Unique}(\mathcal{U})$ 

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2.2 Practical Implementation and Data-Driven Method

In practical applications, both the number and locations of change points $\mathcal{T}_{\kappa^*}^*$ are unknown, and key algorithmic parameters must be specified in a data-driven manner. In this subsection, we first describe the selection of tuning parameters for a fixed initializer count $\tilde{\kappa}$, and then introduce an adaptive algorithm for choosing the number of initializers when the true number of change points is unknown.

Tuning Parameter Selection. For any fixed initializer count $\tilde{\kappa}$, Algorithm 1 involves two tuning parameters, λ and γ , which jointly control the accuracy of coefficient estimation and the complexity of the induced segmentation. To balance computational efficiency and statistical stability, we adopt a simple cross-validation (CV) strategy based on an odd–even data split. Specifically, the data are partitioned into $\mathcal{D}^{\text{odd}}$ and $\mathcal{D}^{\text{even}}$. For each candidate pair (λ, γ) , Algorithm 1 is applied to $\mathcal{D}^{\text{odd}}$, and the resulting segmentation is evaluated on $\mathcal{D}^{\text{even}}$ using a residual sum of squares criterion. The optimal pair $(\lambda_{\diamond}, \gamma_{\diamond})$ is selected by minimizing the CV score; see Algorithm 3 for details. We advocate tuning λ and γ jointly rather than separately, as in Kaul et al. (2019), for three reasons. (1) Although parameter estimation and location search are procedurally separated, they are intrinsically coupled, since the output of the split step directly enters the merge objective. (2) Tuning λ solely for parameter estimation accuracy may not be optimal for location search, because on intervals containing a change point it can favor estimates that approximate weighted averages of segment-specific parameters rather than the true underlying ones (e.g., Cho and Owens, 2024). (3) Tuning γ through a BIC-type criterion requires specifying a penalty constant, which is nontrivial in multiple change-point settings and can materially affect the selected number of change points.

2.2 Practical Implementation and Data-Driven Method

Owing to the low computational cost of our method, a fine-grained grid search over tuning parameters is feasible in practice.

Algorithm 3 Cross-Validation for SaM

Input: Candidate sets $\Lambda, \Gamma, \Lambda_0 \subset \mathbb{R}^+$, initializer count $\tilde{\kappa}$, training data

$\mathcal{D}^{\text{odd}}$, test data $\mathcal{D}^{\text{even}}$.

Output: Selected tuning parameters $(\lambda_\circ(\tilde{\kappa}), \gamma_\circ(\tilde{\kappa}))$.

- 1: Set initializers $\{\tilde{j} \lfloor n_0/(\tilde{\kappa} + 1) \rfloor : \tilde{j} \in [\tilde{\kappa}]\}$.
- 2: **for** each $(\lambda, \gamma) \in \Lambda \times \Gamma$ **do**
- 3: Apply Algorithm 1 to $\mathcal{D}^{\text{odd}}$. Obtain estimators $\hat{\mathcal{T}}_{\tilde{\kappa}^{\text{odd}}}^{\text{odd}}$.
- 4: Compute validation score

$$\text{RSS}(\lambda, \gamma; \tilde{\kappa}) = \min_{\lambda_0 \in \Lambda_0} \sum_{j=1}^{\tilde{\kappa}^{\text{odd}}+1} \sum_{i=\hat{\tau}_{j-1}^{\text{odd}}+1}^{\hat{\tau}_j^{\text{odd}}} \left(y_i^{\text{even}} - \langle \mathbf{X}_i^{\text{even}}, \check{\boldsymbol{\theta}}_{(\hat{\tau}_{j-1}^{\text{odd}}, \hat{\tau}_j^{\text{odd}}]}^{\lambda_0} \rangle \right)^2. \quad (2.9)$$

- 5: **end for**
 - 6: Select $(\lambda_\circ(\tilde{\kappa}), \gamma_\circ(\tilde{\kappa})) = \text{argmin}_{(\lambda, \gamma) \in \Lambda \times \Gamma} \text{RSS}(\lambda, \gamma; \tilde{\kappa})$.
 - 7: Refit Algorithm 1 using $(\lambda_\circ(\tilde{\kappa}), \gamma_\circ(\tilde{\kappa}))$.
 - 8: Rescale estimated change-points to the original scale.
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An Adaptive Version of SaM: SaMA. In practice, the true number of change points κ^* is unknown, and the initializer count $\tilde{\kappa}$ may fail to satisfy $\tilde{\kappa} \geq \kappa^*$. To address this issue, we propose a data-adaptive extension of SaM, termed **SaMA**. The procedure proceeds as follows. (i) Specify a sufficiently large upper bound $\kappa_{\max} \geq 1$ and candidate tuning sets Λ and Γ ; (ii) For each

$\tilde{\kappa} \in [\kappa_{\max}]$, select tuning parameters $(\lambda_{\diamond}(\tilde{\kappa}), \gamma_{\diamond}(\tilde{\kappa}))$ via the CV procedure in Algorithm 3, and compute the corresponding CV score; (iii) Choose

$$\tilde{\kappa}^{\diamond} = \underset{\tilde{\kappa} \in [\kappa_{\max}]}{\operatorname{argmin}} \operatorname{RSS}(\lambda_{\diamond}(\tilde{\kappa}), \gamma_{\diamond}(\tilde{\kappa}); \tilde{\kappa}),$$

and run Algorithm 1 with $\tilde{\kappa}^{\diamond}$.

This design is motivated by the fact that $\tilde{\kappa} < \kappa^*$ typically leads to under-segmentation and large residual error, whereas overly large $\tilde{\kappa}$ results in unstable estimates due to short segments, potentially violating theoretical conditions and degrading localization accuracy.

3. Theoretical Guarantees

For real-world data, requiring an exact low-rank form is often too restrictive, where the rank of each true coefficient matrix $\{\boldsymbol{\theta}_j^*\}_{j=1}^{\kappa^*+1}$ is typically bounded by a small integer. To characterize the low-rank structure of the coefficient matrices, we adopt a near low-rank condition that accommodates both exact and approximate low-rank. Specifically, suppose that each coefficient matrix $\boldsymbol{\theta}_j^*$ belongs to

$$\mathcal{B}(R_{j,q}) = \left\{ \boldsymbol{\theta} \in \mathbb{R}^{d_1 \times d_2} \left| \sum_{l=1}^d [\rho_l(\boldsymbol{\theta})]^q \leq R_{j,q} \right. \right\},$$

where $d = \min\{d_1, d_2\}$, $q \in [0, 1]$ is a relaxation parameter, and $R_{j,q} > 0$ controls the size of the ball. When $q = 0$, this condition reduces to the stan-

standard low-rank constraint $\text{rank}(\boldsymbol{\theta}_j^*) \leq R_{j,0}$. We define $R_q := \max_{j \in [\kappa^*+1]} R_{j,q}$.

3.1 Statistical Assumptions

We first introduce the notion of *index groups*. We consider a grouping structure over the $\tilde{\kappa} + 1$ segments induced by $\tilde{\mathcal{T}}_{\tilde{\kappa}}$. Specifically, under Assumption 1a, which ensures $\tilde{\kappa} \geq \kappa^*$, we partition the index set $[\tilde{\kappa} + 1]$ into $\kappa^* + 1$ non-empty and mutually exclusive *index groups* $\{\mathcal{G}_1, \dots, \mathcal{G}_{\kappa^*+1}\}$ via a marker set $\mathcal{M} := \{m(1), \dots, m(\kappa^*)\} \subseteq \{1, \dots, \tilde{\kappa}\}$. Each group is defined as

$$\mathcal{G}_j = \left\{ \tilde{j} \in [\tilde{\kappa} + 1] \mid m(j-1) < \tilde{j} \leq m(j) \right\}, \quad (3.10)$$

for $j \in [\kappa^* + 1]$, with the convention $m(0) = 0$ and $m(\kappa^* + 1) = \tilde{\kappa} + 1$.

Assumption 1. Define $\{\alpha_n\}_{n \in \mathbb{N}_+}$ as a diverging sequence. There exist constants $\iota \in (0, 1/2]$ and $C_d, C_{\mathcal{R}} > 0$ such that the initializers $\tilde{\mathcal{T}}_{\tilde{\kappa}}$ satisfies the following conditions. (1.a) The number of initializers is no less than the true number of change points, i.e., $\tilde{\kappa} \geq \kappa^*$. (1.b) All initializers are well-separated, $\min_{j \in [\tilde{\kappa}+1]} (\tilde{\tau}_j - \tilde{\tau}_{j-1}) \geq \mathfrak{d}_{\min}$ where $\mathfrak{d}_{\min} \geq C_d \delta_{\min} / \alpha_n^{1-2\iota}$. (1.c) Each true change point has at least one initializer in its fractional neighborhood. That is, for any constant $\tilde{\iota} \in (1 - 2\iota, 1 - \iota]$, there exists a marker set $\mathcal{M}$ such that $\max_{j \in [\kappa^*]} |\tilde{\tau}_{m(j)} - \tau_j^*| \leq \tilde{\mathcal{R}} := C_{\mathcal{R}} \delta_{\min} \Xi_{\min}^{\frac{2}{2-q}} R_q^{-\frac{1}{2-q}} \alpha_n^{-\tilde{\iota}}$.

Remark 1. From the theoretical aspect, in the exact low-rank case $q =$

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0, Assumption (1.c) reduces to $\tilde{\mathcal{R}} = o(n)$ under $\Xi_{\min} R_0^{-1/2} = O(1)$ and $\delta_{\min} = O(n)$. This implies that only one marker initializer is required in a vanishing neighborhood of each true change point. Importantly, we impose no requirements on initializers outside the marker set $\mathcal{M}$. SaM does not select κ^* initializers directly from $\{\tilde{\tau}_j\}_{j=1}^{\tilde{\kappa}}$ as the final estimates; such approaches are closer to BIC-type model selection. Instead, SaM uses the $\tilde{\kappa} + 1$ coefficient estimates only to guide a global search over the full grid $[n]$ and obtain $\hat{\mathcal{T}}_{\tilde{\kappa}}$. In practice, we do not know which initializers are marker initializers; the marker set is only a latent device used in the theoretical analysis. From a practical aspect, a simple way to satisfy Assumption 1.c is to construct a coarse grid. For example, one may choose a slowly diverging sequence $G = n^w$ ($0 < w < 1$), and partition $(0, n]$ into G equally spaced intervals, with $G \geq \kappa^* + 1$. By the pigeonhole principle, at least one initializer must lie within an $O(n^{1-w}) = o(n)$ neighborhood of each true change point. From Table 1, we see that $\tilde{\kappa} = 3$ (i.e., $\tilde{\mathcal{T}}_3 = \{0.25n, 0.5n, 0.75n\}$) already performs well, suggesting that the required initialization radius is about $\tilde{\mathcal{R}} = 0.1n$, which is not restrictive.

A key ingredient of our analysis is a decomposition of the parameter space that exploits the low-rank structure of the matrices within each stationary segment. Recalling the trace regression model (1.1), fix a segment

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j and let $\boldsymbol{\theta}_j^* = \sum_l \sigma_{jl} \mathbf{u}_{jl} \mathbf{v}_{jl}^\top$ be its singular value decomposition. For any integer $r \in [d]$, define $\mathbf{U}_j^r = [\mathbf{u}_{j1}, \dots, \mathbf{u}_{jr}]$ and $\mathbf{V}_j^r = [\mathbf{v}_{j1}, \dots, \mathbf{v}_{jr}]$, and denote by $\mathbf{P}_{\mathbf{U}_j^r}$ and $\mathbf{P}_{\mathbf{V}_j^r}$ the orthogonal projections onto $\mathbf{U}_j^r$ and $\mathbf{V}_j^r$, respectively. For any matrix $\mathbf{W} \in \mathbb{R}^{d_1 \times d_2}$, we define the projection operators

$$\Pi_{\boldsymbol{\theta}_j^*}^r(\mathbf{W}) = \mathbf{P}_{\mathbf{U}_j^r} \mathbf{W} + \mathbf{W} \mathbf{P}_{\mathbf{V}_j^r} - \mathbf{P}_{\mathbf{U}_j^r} \mathbf{W} \mathbf{P}_{\mathbf{V}_j^r},$$

and

$$\Pi_{\boldsymbol{\theta}_j^*}^{r\perp}(\mathbf{W}) = (\mathbf{I}_{d_1} - \mathbf{P}_{\mathbf{U}_j^r}) \mathbf{W} (\mathbf{I}_{d_2} - \mathbf{P}_{\mathbf{V}_j^r}),$$

which will be used to decompose the estimation error into low-rank and orthogonal components within each segment.

Assumption 2. Consider the high-dimensional mean model with change points specified in (1.1). For any $j \in [\kappa^*]$, where $\kappa^* \geq 1$, denote the j -th jump vector by $\boldsymbol{\Psi}_j^* = \boldsymbol{\theta}_{j+1}^* - \boldsymbol{\theta}_j^*$. We further assume that the model (1.1) satisfies the following conditions: (2.a) The design vectors $\{\text{vec}(\mathbf{X}_i)\}_{i=1}^N$ are i.i.d. mean-zero sub-Gaussian with $\sigma_x^2 = \max_i \|\text{vec}(\mathbf{X}_i)\|_{\psi_2}$. Denote the covariance $\boldsymbol{\Sigma} = \mathbb{E}[\text{vec}(\mathbf{X}_i) \text{vec}(\mathbf{X}_i)^\top]$ and its eigenvalues in $[\underline{\rho}_x, \bar{\rho}_x]$. The noise variables $\{\epsilon_i\}_{i=1}^N$ are independent, conditionally mean-zero ($\mathbb{E}[\epsilon_i | \mathbf{X}_i] = 0$), and sub-Gaussian with $\sigma_\epsilon^2 = \max_i \|\epsilon_i\|_{\psi_2}$. (2.b) $\Xi_{\min} = \min_{j \in [\kappa^*]} \Xi_j = \min_{j \in [\kappa^*]} \|\boldsymbol{\Psi}_j^*\|_F$. In addition, for any $j \in [\kappa^*]$, assume that there exists an absolute constant $C_\Xi > 0$ such that $\Xi_j \leq C_\Xi$.

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Assumption 3. There exists a constant $C_{snr} > 0$ such that: $\delta_{\min} \Xi_{\min}^2 \geq C_{snr} \alpha_n (\kappa^* + 1) \mathfrak{d}_{\min}^{q/2} R_q (d_1 + d_2)^{1-q/2}$.

For the exact low-rank setting ($q = 0$), the right-hand side of the above inequality depends on the number of change points, the maximal rank of the true coefficient matrices, and the problem dimensions. In the near low-rank setting ($q \in (0, 1]$), however, the rank may be large, and we therefore introduce an “effective rank” defined through the threshold $\lambda/(\sqrt{\mathfrak{d}_{\min} \rho_x})$, corresponding to the number of sufficiently large singular values. This choice reflects a tradeoff between estimation and approximation errors. Consequently, Assumption 3 is more restrictive in the near low-rank setting to accommodate the approximation error induced by rank truncation. Based on these assumptions, we establish the following upper bounds for the efficient estimation error.

Theorem 1. *Suppose Assumptions 1-3 hold. Let $\lambda = 2(\lambda' + \lambda'')$ where $\lambda' = C' \sqrt{d_1 + d_2}$ and $\lambda'' = C'' \tilde{\mathcal{R}} / \sqrt{\mathfrak{d}_{\min}}$ for absolute constants $C', C'' > 0$. Then, for any $\tilde{j} \in \mathcal{G}_j$ with $j \in [\kappa^* + 1]$, the following holds with probability at least $1 - o(1)$:*

- (i) $\|\tilde{\boldsymbol{\theta}}_{\tilde{j}} - \boldsymbol{\theta}_j^*\|_F^2 \leq C_1 \lambda^{2-q} R_q / \mathfrak{d}_{\min}^{1-\frac{q}{2}} \leq C_2 \alpha_n^{-2\iota^*} \Xi_{\min}^2$;
- (ii) $\|\tilde{\boldsymbol{\theta}}_{\tilde{j}} - \boldsymbol{\theta}_j^*\|_* \leq 4 \|\Pi_{\boldsymbol{\theta}_j^*}^r(\tilde{\boldsymbol{\theta}}_{\tilde{j}} - \boldsymbol{\theta}_j^*)\|_* + 4 \sum_{l=r+1}^d \rho_l(\boldsymbol{\theta}_j^*) \leq C_3 \lambda^{2-2q} R_q^2 / \mathfrak{d}_{\min}^{1-q}$, for some absolute constants $C_1, C_2, C_3 > 0$ and $\iota^* = \tilde{\iota} - (1 - 2\iota)$.

Theorem 1 provides non-asymptotic error bounds for the matrix estimators in both Frobenius and nuclear norms under near low-rank settings. In the exact low-rank case ($q = 0$), the bounds depend explicitly on the maximal rank parameter R_q of the true coefficient matrices and the problem dimensions d_1, d_2 . By choosing λ of the prescribed order, the resulting estimation errors match the optimal rates established for static low-rank trace regression models without change points (e.g., Negahban and Wainwright, 2011). This indicates that the presence of multiple change points does not incur any additional loss in estimation accuracy for coefficient recovery.

Theorem 2. *Suppose Assumptions 2 and 3 hold, and assume that the initial parameters $\tilde{\boldsymbol{\theta}}_{[\hat{\kappa}+1]}$ satisfy the properties in Theorem 1. Let the estimated change points $\hat{\mathcal{T}}_{\hat{\kappa}}$ be obtained from Algorithm 1 and effective rank $\bar{r}$ defined in Appendix ??, using a tuning parameter γ satisfying $C_\gamma n \lambda^2 \bar{r} / \mathfrak{d}_{\min} \leq \gamma \leq c_\gamma \delta_{\min} \Xi_{\min}^2$, for some constants $C_\gamma, c_\gamma > 0$. Then, the following results hold:*

- (i) $\mathbb{P}\{\hat{\kappa} = \kappa^*\} \rightarrow 1$, as $n, d \rightarrow \infty$;
- (ii) $\max_{1 \leq j \leq \hat{\kappa}} |\hat{\tau}_j - \tau_j^*| = O_p(\lambda^{2-q} R_q / (n^{-q/2} \Xi_{\min}^2))$.

Theorem 2 establishes consistency and localization guarantees for the proposed **SaM** algorithm with multiple change points. Part (i) shows that the estimated number of change points $\hat{\kappa}$ is selection consistent, recovering the true κ^* with probability tending to one as $n, d \rightarrow \infty$, provided that

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the tuning parameter γ lies in the prescribed range. Part (ii) characterizes the localization accuracy, showing that the maximal estimation error over all detected change points is controlled by the coefficient estimation error. Specifically, the localization rate depends on $\lambda^{2-q}R_q$ and scales inversely with the effective signal strength $n^{-q/2}\Xi_{\min}^2$, highlighting the trade-off between low-rank approximation error and change-point separability. Based on the definition of the effective-rank upper bound $\bar{r}$, the localization accuracy can be expressed as $O_p(\bar{r}(d_1 + d_2)/\Xi_{\kappa^*}^2)$ (see (??)). This rate is broadly consistent with the theoretical localization results for change-point estimation in linear regression models (Rinaldo et al., 2021). In the exact low-rank case ($q = 0$), the bound reduces to the standard parametric rate determined by the true rank and signal strength, recovering known optimal localization results (e.g., Shi et al., 2024).

Why SaM Works. By construction of (3.10), each group contains at least one index. Consequently, the initial coefficient estimators $\tilde{\boldsymbol{\theta}}_{[\tilde{\kappa}+1]}$ can be grouped accordingly from Theorem 2(i), under which the convergence rate may be arbitrarily slow. We briefly illustrate why (2.5) leads to effective change-point detection using the illustrative example in Figure 1. Consider a simple case where an index group is $\mathcal{G}_2 = \{3, 4\}$. By the construction of the total loss in (2.4), two local losses are involved: (i) $\mathcal{L}(\hat{u}_2, \hat{u}_3; \tilde{\boldsymbol{\theta}}_3)$

and (ii) $\mathcal{L}(\hat{u}_3, \hat{u}_4; \tilde{\boldsymbol{\theta}}_4)$. These losses are simultaneously small only when the segments align with the true interval $(\tau_1^*, \tau_2^*]$, as implied by $\|\tilde{\boldsymbol{\theta}}_{\tilde{j}} - \boldsymbol{\theta}_2^*\|_F = O_p(\alpha_n^{-2\iota^*} \Xi_{\min}^2)$, for $\tilde{j} \in \mathcal{G}_2$, and $\mathcal{D}_{\tau_1^*}^{\tau_2^*} \sim \mathbb{P}(\boldsymbol{\theta}_2^*)$. With a sufficiently large penalty γ , the optimization favors $\hat{u}_3 = \hat{u}_4$, effectively merging the two locations and eliminating loss (ii). As a result, each index group contributes exactly one active segment. The same argument applies to other groups, such as $\mathcal{G}_1 = \{1, 2\}$. Consequently, after deduplication, the number of effective segments induced by $\hat{\mathcal{U}}_{\hat{\kappa}}$ converges to $\kappa^* + 1$.

4. Simulation and Real Data Applications

In this section, we assess the performance of the proposed method on synthetic data and demonstrate its utility through real data applications.

4.1 Simulation

Evaluation Metrics. We first evaluate the statistical efficiency of the proposed method. To assess the accuracy of estimating the number of change points, we report the mean estimated number $\mathbb{E}[\hat{\kappa}]$ computed over 50 replications, together with the overestimation and underestimation frequencies $\mathbb{P}(\hat{\kappa} > \kappa^*)$ and $\mathbb{P}(\hat{\kappa} < \kappa^*)$, respectively. We further evaluate the localization accuracy of the change point estimators $\hat{\mathcal{T}}_{\hat{\kappa}}$ using the standardized Haus-

distance d_H . Specifically, we report the mean of d_H , $\mathbb{E}[d_H]$, and its standard deviation $\mathbf{Std}(d_H)$ (in parentheses). The standardized Hausdorff distance is defined as

$$d_H = \frac{1}{n} \max \left\{ \max_{j_1=0, \dots, \widehat{\kappa}+1} \min_{j_2=0, \dots, \kappa^*+1} |\widehat{\tau}_{j_1} - \tau_{j_2}^*|, \max_{j_2=0, \dots, \kappa^*+1} \min_{j_1=0, \dots, \widehat{\kappa}+1} |\widehat{\tau}_{j_1} - \tau_{j_2}^*| \right\},$$

with boundary points defined as $\widehat{\tau}_0 = 0$ and $\widehat{\tau}_{\widehat{\kappa}+1} = n$.

Settings. We conduct Monte Carlo experiments to evaluate the empirical performance of the proposed **SaMA** method, which selects the number of initializers in a fully data-driven manner as described in Section 2.2. Throughout the experiments, the true number of change points is fixed at $\kappa^* = 2$ with locations $\tau_1^* = 0.25n$ and $\tau_2^* = 0.6n$, and the candidate number of initializers $\widetilde{\kappa}$ ranges over $\{1, \dots, 8\}$ with equally spaced initializers. We focus on independent designs in the main text, where the design matrices $\{\mathbf{X}_i\}_{i=1}^n$ have i.i.d. standard normal entries and the noise variables follow $\varepsilon_i \sim \mathcal{N}(0, 0.1^2)$. The true coefficient matrices are low-rank with fixed rank $r^* = 5$ and vary across segments, inducing structural changes between regimes. We first provide detailed results for the baseline setting with $d = 20$ and $n = 2000$ in Table 1, followed by additional results across sample sizes in Table 2. Moreover, the details for true coefficient matrix generation, additional simulation results with $d = 40$, smaller jump sizes, and *temporal dependence* are deferred to the Appendix ??.

Table 1: Change-point detection performance with $d = 20$ and $n = 2000$ under independent designs.

Methods	$\mathbb{E}[\hat{\kappa}]$	$\mathbb{P}(\hat{\kappa} > \kappa^*)$	$\mathbb{P}(\hat{\kappa} < \kappa^*)$	$\mathbb{E}[d_H]$	$\text{Std}(d_H)$
$\tilde{\kappa} = 1$	1.000	0.000	1.000	0.256 0.256	(0.010 0.011)
$\tilde{\kappa} = 2$	2.000	0.000	0.000	0.081 0.000	(0.006 0.000)
$\tilde{\kappa} = 3$	2.000	0.000	0.000	0.000 0.000	(0.000 0.000)
$\tilde{\kappa} = 4$	2.740	0.740	0.000	0.111 0.107	(0.066 0.067)
$\tilde{\kappa} = 5$	2.120	0.060	0.000	0.006 0.004	(0.024 0.020)
$\tilde{\kappa} = 6$	2.460	0.440	0.000	0.068 0.028	(0.068 0.079)
$\tilde{\kappa} = 7$	2.000	0.000	0.000	0.001 0.000	(0.001 0.000)
$\tilde{\kappa} = 8$	3.300	0.680	0.000	0.061 0.046	(0.033 0.041)
SaMA	2.020	0.020	0.000	0.004 0.003	(0.021 0.021)
Shi et al. (2024)	2.000	0.000	0.000	0.025	(0.022)

Table 1 summarizes the change-point detection results. For the rows corresponding to **SaM** variants, the Hausdorff columns report preliminary and refined results separated by “|”. When the number of initializers is severely under-specified, the method underestimates the number of change points and suffers from large localization errors. With sufficiently many initializers, accurate recovery becomes possible, while too many initializers may introduce additional variability. The proposed **SaMA** method effectively removes the dependence on a fixed choice of $\tilde{\kappa}$, selecting an average of 3.48

Table 2: Change-point detection performance across different sample sizes with $d = 20$ under independent designs.

Setting	Method	$\mathbb{E}[\hat{\kappa}]$	$\mathbb{P}(\hat{\kappa} > \kappa^*)$	$\mathbb{P}(\hat{\kappa} < \kappa^*)$	$\mathbb{E}[d_H]$	$\text{Std}(d_H)$
$n = 1500$	SaMA	2.240	0.240	0.000	0.035	0.063
	Shi et al. (2024)	2.000	0.000	0.000	0.007	0.002
$n = 1750$	SaMA	2.180	0.140	0.000	0.011	0.038
	Shi et al. (2024)	2.000	0.000	0.000	0.007	0.003
$n = 2000$	SaMA	2.020	0.020	0.000	0.003	0.021
	Shi et al. (2024)	2.000	0.000	0.000	0.025	0.022
$n = 2250$	SaMA	2.000	0.000	0.000	0.000	0.000
	Shi et al. (2024)	2.000	0.000	0.000	0.071	0.035
$n = 2500$	SaMA	2.000	0.000	0.000	0.000	0.000
	Shi et al. (2024)	2.000	0.000	0.000	0.001	0.001

initializers in this setting while consistently recovering the correct number of change points and achieving near-zero Hausdorff distances. The refined estimates further improve upon the preliminary localizers in several over-specified settings.

To examine the stability of the methods beyond the baseline setting, we further report results across different sample sizes with $d = 20$ in Table 2. The proposed **SaMA** method exhibits a clear improvement as n increases.

The mean estimated number of change points, $\mathbb{E}[\hat{\kappa}]$, decreases from 2.24 at $n = 1500$ to the true value 2.00 when $n \geq 2250$, while both the mean and standard deviation of the Hausdorff distance converge to zero. The method of Shi et al. (2024) also recovers the correct number of change points across all settings, but its localization performance is sensitive to tuning parameters. Since its runtime is substantially longer, as shown in Figure 2, an exhaustive tuning search is difficult in practice.

Given the comparable statistical accuracy observed above, we next compare the computational costs in Figure 2, which reports runtime in seconds on a logarithmic scale for $d = 20$ and $d = 40$. Across both dimensions, **SaMA** is much faster than Shi et al. (2024), and its runtime is only mildly affected by the increase in n . In contrast, the runtime of Shi et al. (2024) grows markedly with the sample size, making extensive tuning difficult in larger samples. This computational stability reflects the split-and-merge construction: **SaMA** decouples coefficient estimation from change-point localization and avoids repeatedly re-estimating low-rank coefficient matrices over many candidate segments.

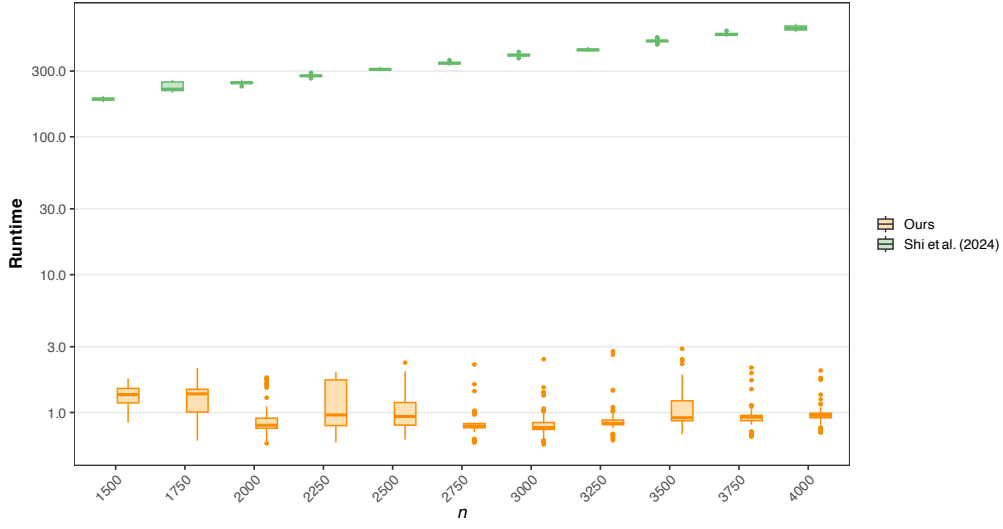
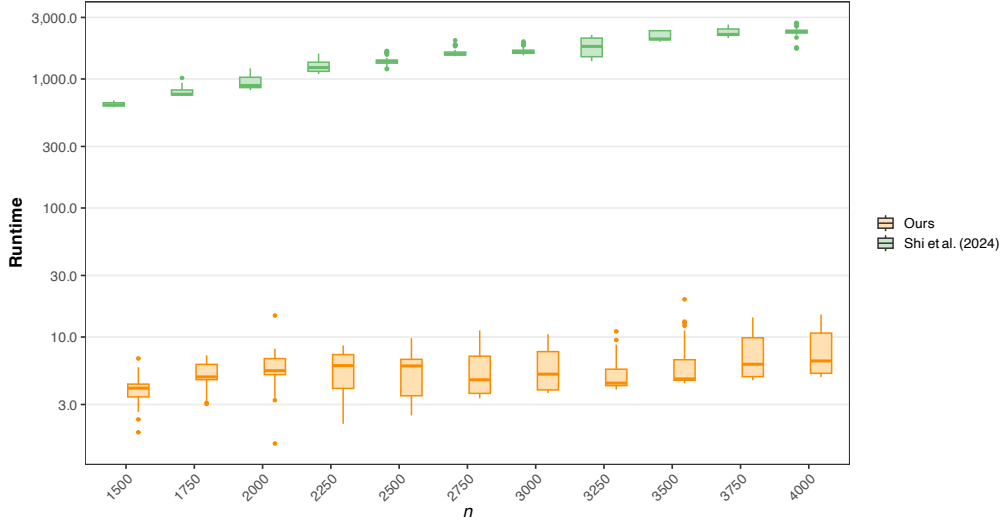
(a) $d = 20$ (b) $d = 40$

Figure 2: Runtime comparison with varying sample size n under independent designs. The vertical axis reports runtime in seconds on a logarithmic scale.

4.2 Real Data Applications

In this section, we apply **SaMA** to two real-world datasets, the Beijing air-quality data and the Fama–French 49 industry portfolio data, to illustrate its empirical performance and interpretability in environmental and financial applications.

Air-Quality Data. We apply the proposed **SaMA** procedure to an hourly air-quality dataset collected from 12 nationally controlled monitoring sites operated by the Beijing Municipal Environmental Monitoring Center, covering the period from March 1, 2013 to February 28, 2017. The dataset, which records concentrations of SO_2 , CO , NO_2 , and O_3 , is publicly available from the UCI Machine Learning Repository (Chen, 2017). We use the daily average $\text{PM}_{2.5}$ across all stations as the response, and organize the pollutant measurements as matrix-valued predictors indexed by monitoring site and pollutant type. The **SaMA** algorithm detects three change points at February 16, 2015, November 15, 2015, and December 24, 2016, where the first and third are highly consistent with those reported in Shi et al. (2024). Based on these change points, we estimate segment-wise coefficient matrices and visualize their induced station-level effects in Figure 3, where bubble sizes represent the station-wise aggregated magnitude of the estimated coefficients weighted by the segment-wise average covariates, and

colors indicate the dominant contributing pollutant.



Figure 3: Structural changes in station-level pollutant effects across detected segments. Panels (a)–(d) display the four segments obtained by partitioning the data at the estimated change points.

Figure 3 reveals clear structural changes across segments. After the first change point, urban SO₂-related effects decrease markedly. Following the second change point, effects across stations become uniformly weaker,

suggesting an overall improvement in air quality. In the final segment, which overlaps with the winter heating season, urban stations exhibit increased effects, likely reflecting seasonal heating demand.

Fama–French 49 Data. We further evaluate the proposed **SaMA** method on the Fama–French 49 industry portfolio data from 2005 to 2015, obtained via the `frenchdata` R package. At each trading day t , the response variable is taken as the absolute market excess return $|\text{Mkt}_t - \text{RF}_t|$, serving as a proxy for overall market volatility, while the matrix-valued predictor is constructed as the rolling correlation matrix of the 49 industry returns over a 60-day window prior to t . Applying **SaMA** to this dataset, we detect three change points at July 23, 2007, December 8, 2008, and June 7, 2010, which partition the sample into four segments roughly corresponding to the pre-crisis period, the onset of the global financial crisis, the crisis aftermath, and the subsequent recovery phase. As a comparison, applying the method of Shi et al. (2024) to the same dataset does not detect any change point, suggesting that the financial example is challenging for procedures that rely on joint stacked estimation of all segment-specific parameters. Figure 4 visualizes the estimated coefficient matrices for the segments detected by **SaMA** and reveals pronounced structural differences across time. In the first segment, the coefficient matrix exhibits a clear cross-shaped pattern, indicating

that a subset of industries plays a dominant role in linking industry-level dependence to market volatility. During the financial crisis period, coefficients become large in magnitude but highly irregular. In the post-crisis segment, the patterns remain heterogeneous but start to weaken. In the final segment, the coefficients are smaller and more homogeneous overall, suggesting a normalization in the relationship between industry dependence and market volatility.

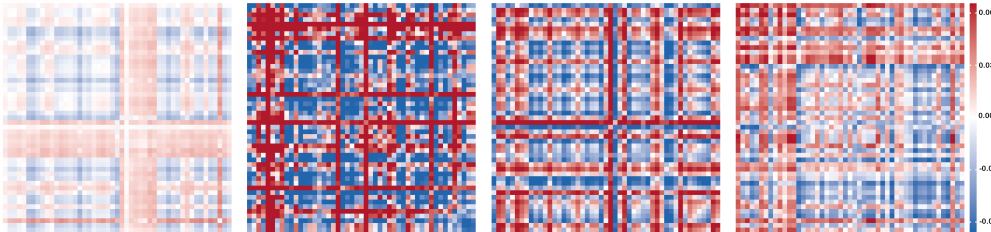


Figure 4: Structural changes in estimated coefficient matrices across detected segments for the Fama–French 49 data.

5. Conclusion

We considered multiple change-point detection in dynamic low-rank trace regression models and proposed a **SaM** framework that decouples low-rank parameter estimation from change-point localization. This design achieves near-optimal localization accuracy while reducing the computa-

tional cost from multiplicative to additive, enabling sub-second performance in high-dimensional regimes. Extensive simulations and real-data studies demonstrate that **SaM** is both statistically competitive and computationally efficient. Future work includes extending the framework to online settings and more general forms of time-varying low-rank structures.

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Supplementary Materials

The supplementary materials provide additional numerical results, supplementary algorithmic details, the global refinement procedure, and detailed proofs of the main theorems.

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